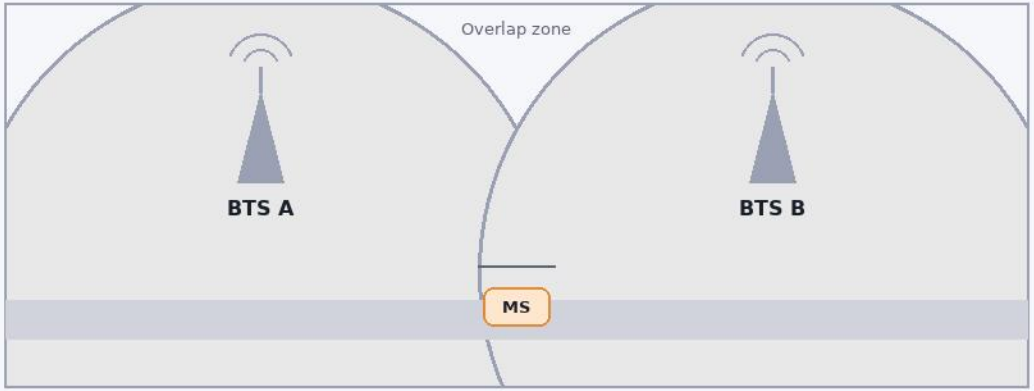
Ex-6 DESIGN: Handoff scenario layout – cells, base stations and mobile station



Design specifications

Frame: Desktop 1440 x 800

Cells: Ellipse, 460 x 460, overlap 60 px

BTS: Triangle + arc icons (signal)

MS: Rounded rectangle on road line

Signal bars: Rectangle, width = strength

Link line: 3 px (purple A, green B)

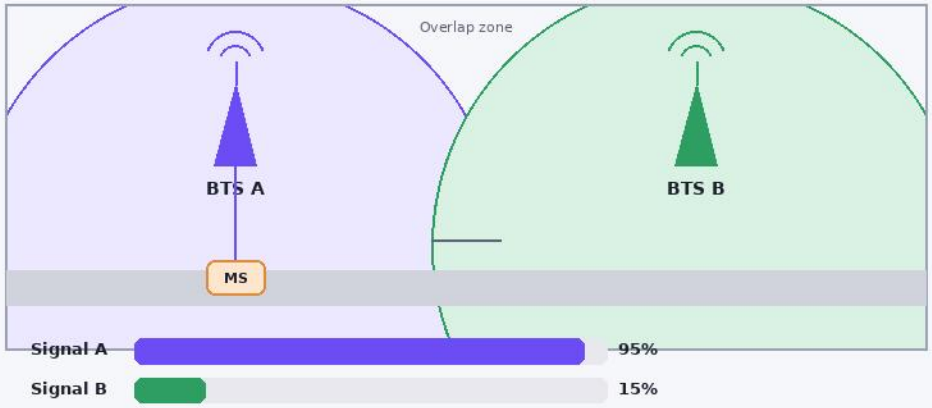
Handoff command: red line + label

Font: Inter Semi Bold 16

Elements: BTS A, BTS B, Overlap zone, MS, Road, Signal meters A and B

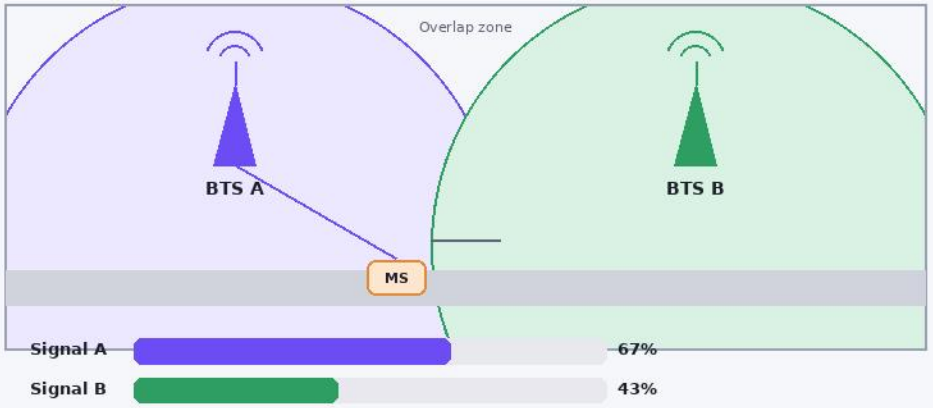
Ex-6 PROTOTYPE: Handoff simulation (On click → Navigate to, Smart Animate)

Frame 1: MS connected to BTS A



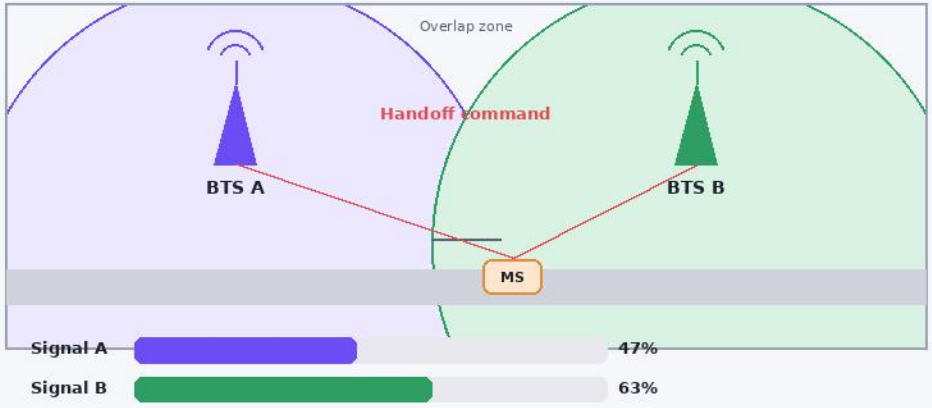
Strong signal from A, B not detected

Frame 2: MS moves, A weakens



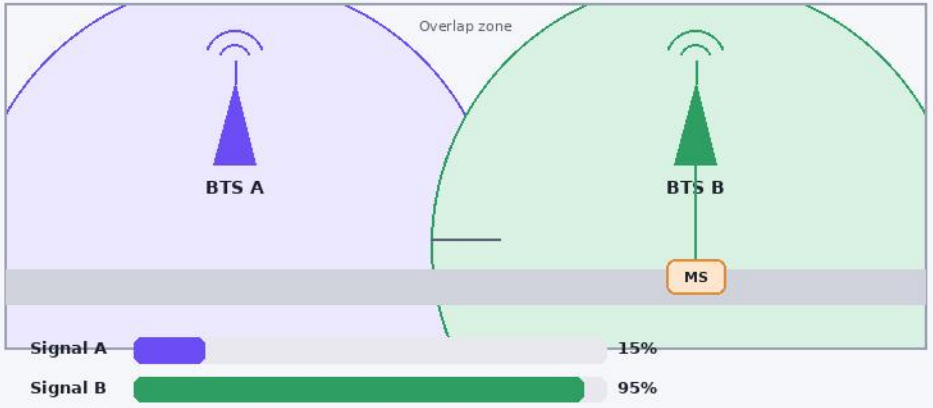
Measurement report sent to BSC

Frame 3: Handoff executing



B stronger by margin → command to MS

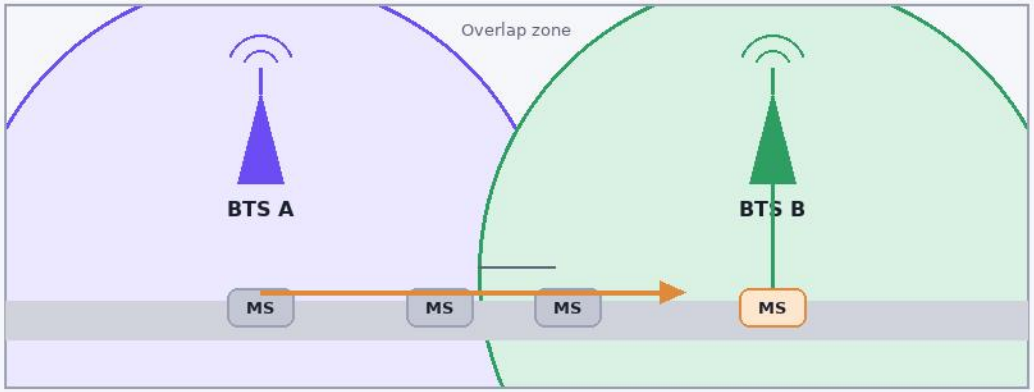
Frame 4: Connected to BTS B

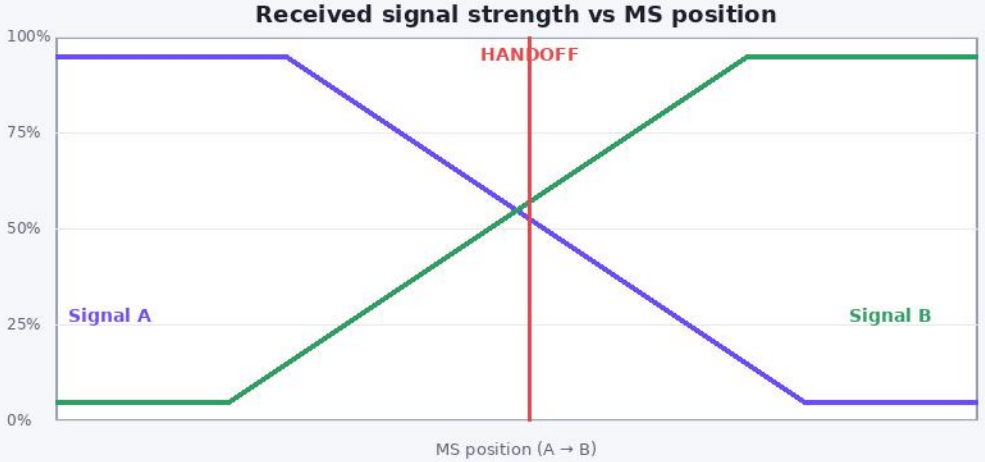


New channel active, A channel released

Trigger: On click “Move MS” (or After delay 1 s) | Action: Navigate to | Animation: Smart animate, 600 ms | Start frame: Frame 1

Ex-6 OUTPUT: Handoff scenario – MS moves from BTS A to BTS B





Handoff process

1. MS is connected to BTS A (strong signal)
2. MS measures A and neighbour B, reports to BSC
3. BSC compares: B stronger than A by margin
4. BSC sends handoff command to MS and BTS B
5. MS tunes to the new channel of BTS B
6. Call continues on B; channel on A released

Result: seamless hard handoff, call not dropped.
Hysteresis margin prevents ping-pong handoffs.